CAESAR software architecture

Model registration
UFL_MODELS → CAESAR catalog
Registration + checkpoint SHA-256

Export and installation
3 packages + 6 CDF tables
Manifest: identity, ranks, device

Native model loading
Validate installation manifest
Thread-local runners / tables

COMPRESSION →

Input tensor
Values and shape
Error bound

Preprocessing
Normalize and block
Rank; padding / filtering

Neural encoding
Analysis + hyperprior
Quantized latents

rANS encoding
Latents / hyperlatents
Installed CDF tables

Original tensor reference

Quantized latents

Reconstruction metadata
Model ID; shape and rank
Normalization; blocks, padding, filtering
Correction method + metadata

Local reconstruction
Hyperprior + synthesis
Restore blocks and scale

Residual correction
Input vs. reconstruction
GAE / LBRC / NGLR

Serialized compressed representation
Model identity + metadata + latent streams + correction payload (NGLR weights when used)

← DECOMPRESSION

Parse and validate
Streams and metadata
Check installed model ID

rANS decoding
Recover latent symbols
Installed CDF tables

Reconstruction
Hyperprior + synthesis
Restore blocks and scale

Apply correction
GAE / LBRC / NGLR
Reconstructed tensor

Blue: neural Orange: correction Green: rANS White: data / metadata